Microeconomics I — Class 9: key

Practice for the Part I exam · 27 October 2026 · Pavel Kocourek

Final answers only, to check your own work; every step and every reason is in the *worked solutions*. CV and EV with MWG's signs (negative = a loss): $\mathrm{CV} = w - e(p^1, u^0)$, $\mathrm{EV} = e(p^0, u^1) - w$.

Problem 1 · $u = x_1 x_2^2$; $p^0 = (1,1) \to p^1 = (1,8)$, $w = 6$

classes 4–7

(a) $x_1(p,w) = \dfrac{w}{3p_1}$, $x_2(p,w) = \dfrac{2w}{3p_2}$; at p^0, $w = 6$: $x = (2,4)$

(b) $v(p,w) = \dfrac{4w^3}{27\,p_1 p_2^2}$; $u^0 = v(p^0, 6) = 32$

(c) $e(p,u) = 3\Big(\dfrac{p_1 p_2^2\, u}{4}\Big)^{1/3}$

(d) $h_1(p,u) = \dfrac{e(p,u)}{3p_1}$, $h_2(p,u) = \dfrac{2\,e(p,u)}{3p_2}$; this is $x\big(p, e(p,u)\big)$

(e) $x(p^1, w) = (2, \frac{1}{2})$, $u^1 = \frac{1}{2}$; $e(p^1, u^0) = 24$, $h(p^1, u^0) = (8,2)$
$\Delta x_2 = -\frac{7}{2}$: substitution effect -2, wealth effect $-\frac{3}{2}$
$h_1(p^1, u^0) = 8 > x_1(p^0, w) = 2$: compensated, she substitutes towards good 1 ($x_1(p^1, w)$ stays 2)
$\mathrm{CV} = -18$; $e(p^0, u^1) = \frac{3}{2}$, $\mathrm{EV} = -\frac{9}{2}$

Problem 2 · $u = 2\sqrt{x_1} + 3\sqrt{x_2}$; $p^0 = (1,3) \to p^1 = (8,3)$, $w = 56$

classes 4–7

(a) $x_1(p,w) = \dfrac{4w\,p_2}{p_1(4p_2 + 9p_1)}$, $x_2(p,w) = \dfrac{9w\,p_1}{p_2(4p_2 + 9p_1)}$; at p^0, $w = 56$: $x = (32,8)$
interior: marginal utilities $\to \infty$ as $x_i \to 0$; FOCs sufficient: u concave, budget set convex

(b) $v(p,w) = \sqrt{\dfrac{w\,(4p_2 + 9p_1)}{p_1 p_2}}$; $u^0 = 14\sqrt{2}$, $(u^0)^2 = 392$

(c) $e(p,u) = \dfrac{u^2\, p_1 p_2}{4p_2 + 9p_1}$

(d) $h_1(p,u) = \dfrac{4u^2\, p_2^2}{(4p_2 + 9p_1)^2}$, $h_2(p,u) = \dfrac{9u^2\, p_1^2}{(4p_2 + 9p_1)^2}$

(e) $x(p^1, w) = (1,16)$, $u^1 = 14$; $e(p^1, u^0) = 112$, $\mathrm{CV} = -56$; $e(p^0, u^1) = 28$, $\mathrm{EV} = -28$
$h(p^1, u^0) = (2,32)$; $\Delta x_1 = -31$: substitution effect -30, wealth effect -1

Problem 3 · Martina, $u(x) = \sqrt{x}$; the die game

class 8

(a) prizes $0, 9, 81, 144$ euros with probabilities $\frac{1}{4}, \frac{1}{3}, \frac{1}{3}, \frac{1}{12}$

(b) $E[x] = 42$ euros, $E[u(x)] = 5$, $c = 25$ euros, $\mathrm{RP} = 17$ euros; u strictly concave: risk averse, $c < E[x]$, $\mathrm{RP} > 0$

(c) independence: the common branch (A with probability $\frac{1}{3}$) does not matter; she compares s with B, $c_B = 36$: $s \geq 36$ euros

(d) Pau (risk neutral): $s \geq 45$ euros ($= E_B[x]$); a strict risk lover needs a larger s: more than 45 euros

Problem 4 (optional) · $u = x_1 + \min\{x_1, x_2\}$ on $\mathbb{R}^2_+$

class 2

(a) monotone: **yes** ($y \gg x$ raises both terms) **(b)** strongly monotone: **no** ($u(1,3) = u(1,2) = 2$)

(c) LNS: **yes** (a little more of good 1 is better) **(d)** convex: **yes** ($u = \min\{2x_1,\ x_1 + x_2\}$ is concave)